

Quantile Regression Analysis of sPCI-pPCI Discrepancy

Notebook: 02-assoc_difference_quantile.ipynb

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Outcome: raw_diff = sPCI - pPCI (surgical minus pathological PCI score)

1. Study Sample

Statistic	Value
N	412
Mean raw_diff	+7.10
Median raw_diff	+5.00
SD	7.00
Range	-8 to +32

The distribution is **right-skewed** (mean > median), with a long upper tail of large overestimates. Quantile regression is used instead of OLS to model the conditional distribution without normality assumptions.

Tumor type 1 (n = 122, the most frequent) is the reference category. Tumor type 8 was excluded. Final predictors: Sex, Age, Zn_HIPEC, PreOP_CTx, Thermoablation, and dummy variables for tumor types 2-7.

2. Full Model (including Age)

2.1 Median Model (q = 0.50)

Predictor	Coef	p	Bootstrap 95% CI
Intercept	+6.72	0.007	[+2.79, +12.50] *
Sex	-0.48	0.500	[-2.00, +0.98]
Age	-0.04	0.116	[-0.10, +0.01]
Zn_HIPEC	+2.98	0.076	[+0.28, +4.89] *
PreOP_CTx	-1.32	0.119	[-3.29, -0.00] *
Thermoablation	-2.15	0.005	[-4.46, -1.00] *
Tumor_type_2	-0.05	0.974	[-2.79, +2.81]
Tumor_type_3	-0.06	0.963	[-2.54, +4.47]
Tumor_type_4	+0.80	0.440	[-1.37, +3.00]
Tumor_type_5	+5.87	<0.001	[+3.17, +8.00] *
Tumor_type_6	-1.67	0.157	[-4.00, -0.00] *
Tumor_type_7	+6.68	<0.001	[+2.61, +9.43] *

2.2 Quantile Process Summary

Predictor	q=0.25	q=0.50	q=0.75	q=0.90
Sex	+0.20	−0.48	−2.00	−4.00 *
Age	−0.02	−0.04	0.00	0.00
Zn_HIPEC	+1.77	+2.98	+5.00 *	0.00
PreOP_CT _x	−0.65	−1.32	−2.00	−1.21
Thermoablation	−2.68 ***	−2.15 **	−2.00	−4.00 *
Tumor_type_5	+3.83 ***	+5.87 ***	+11.00 ***	+13.00 ***
Tumor_type_7	+4.26 ***	+6.68 ***	+6.00 **	+8.00 **

(All other tumor types non-significant at all quantiles.)

2.3 Model Fit

Quantile	Pseudo-R ²
q=0.25	0.087
q=0.50	0.127
q=0.75	0.156
q=0.90	0.215
OLS R ²	0.224

2.4 Symmetry Test (q=0.75 – q=0.25, bootstrap)

Predictor	Δ coef	95% CI	Asymmetric?
Sex	−2.24	[−5.00, −0.00]	YES *
Tumor_type_5	+5.92	[+1.55, +10.72]	YES *
All others	—	CI includes 0	No

3. Reduced Model (Age Excluded)

Age was **not significant at any quantile** (all $p > 0.10$, bootstrap CI includes zero). The analysis was repeated without Age.

3.1 Median Model (q = 0.50)

Predictor	Coef	p	Bootstrap 95% CI
Intercept	+4.00	0.043	[+2.00, +9.00] *
Sex	−1.00	0.162	[−2.00, +1.00]
Zn_HIPEC	+3.00	0.074	[+0.37, +5.00] *

Predictor	Coef	p	Bootstrap 95% CI
PreOP_CTx	−1.00	0.236	[−3.00, +0.00]
Thermoablation	−2.00	0.009	[−5.00, −1.00] *
Tumor_type_2	0.00	1.000	[−3.00, +3.00]
Tumor_type_3	0.00	1.000	[−3.00, +4.99]
Tumor_type_4	+1.00	0.333	[−1.19, +3.00]
Tumor_type_5	+6.00	<0.001	[+3.39, +8.00] *
Tumor_type_6	−1.00	0.394	[−4.00, +0.00]
Tumor_type_7	+7.00	<0.001	[+3.00, +9.77] *

3.2 Quantile Process Summary

Predictor	q=0.25	q=0.50	q=0.75	q=0.90
Sex	+0.33	−1.00	−2.00	−4.00 *
Zn_HIPEC	+1.67	+3.00	+5.00 *	0.00
PreOP_CTx	−0.67	−1.00	−2.00	−1.14
Thermoablation	−2.67 ***	−2.00 **	−2.00	−4.00 *
Tumor_type_4	−0.33	+1.00	+2.00	+4.87 *
Tumor_type_5	+3.67 ***	+6.00 ***	+11.00 ***	+13.00 ***
Tumor_type_7	+4.00 ***	+7.00 ***	+6.00 **	+8.00 **

3.3 Model Fit

Quantile	Pseudo-R ²
q=0.25	0.086
q=0.50	0.124
q=0.75	0.156
q=0.90	0.215
OLS R ²	0.223

Removing Age reduces OLS R² by only 0.001 (0.224 → 0.223), confirming it contributes nothing.

3.4 Symmetry Test (q=0.75 – q=0.25, bootstrap)

Predictor	Δ coef	95% CI	Asymmetric?
Intercept	+5.65	[+0.50, +14.04]	YES *
Sex	−2.28	[−5.00, −0.00]	YES *
Tumor_type_4	+2.63	[+0.00, +5.94]	YES *
Tumor_type_5	+6.03	[+1.63, +11.00]	YES *
All others	—	CI includes 0	No

4. Diagnostics

All four quantile models passed all diagnostic checks:

- **Residual sign proportions** match the nominal quantile at all q ($\pm 5\%$ tolerance, $\square$), indicating numerical convergence.
 - **Residuals vs fitted** show no systematic trend or heteroskedasticity.
 - **Spearman correlations** between residuals and all predictors are near zero and non-significant, ruling out missed non-linearity or obvious omitted variables.
 - **Bootstrap SEs vs asymptotic SEs** agree closely for all significant predictors, validating the sparsity estimator.
 - IterationLimitWarning messages arise only in a small number of edge-case bootstrap resamples, not in the primary fitted models; they do not affect the conclusions.
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5. Conclusions

Primary finding

Tumor type drives the sPCI-pPCI discrepancy. Patients with tumor type 5 have a median overestimate roughly 6 points higher than type 1, growing to +11 at $q=0.75$ and +13 at $q=0.90$. Tumor type 7 shows a similar pattern (+7 at the median, +6-8 in the upper quartiles). All other tumour types are indistinguishable from the reference.

Thermoablation reduces the discrepancy

Thermoablation is consistently associated with a **2-4 point reduction** in sPCI-pPCI across all quantiles. The effect is most pronounced at the lower and upper tails ($q=0.25$ and $q=0.90$).

Asymmetric effects

The symmetry test flags two variables whose effects **differ between the lower and upper halves of the distribution**:

- **Sex:** the negative effect of being male appears only in high-discrepancy patients (upper tail).
- **Tumor_type_5:** the overestimation effect is substantially larger in patients who already have a large discrepancy, indicating that surgeons are disproportionately wrong in the worst cases.

In the reduced model, **Tumor_type_4** also shows borderline asymmetry emerging only at $q=0.90$, which was not apparent when Age was included.

Age and other predictors

Age, PreOP_CTx, Zn_HIPEC (at most quantiles), and tumour types 2, 3, and 6 have no robust association with the discrepancy. Dropping Age has negligible impact on model fit ($\Delta\text{OLS } R^2 < 0.001$).

Model fit

The pseudo- R^2 increases from ~ 0.086 at $q=0.25$ to ~ 0.215 at $q=0.90$, closely matching the OLS R^2 (0.22). This confirms that the predictors — particularly tumor type — account for more variance in the upper tail, where discrepancies are largest and clinically most consequential.